Lecture 15

02 March 2026 16:22

Recap

$\min_{x \in \mathbb{R}^n} f(x)$

Newton's method

Initialize: $x_0 \in \mathbb{R}^n$

Iterate: for $t \geq 0$, $x_{t+1} = x_t - \nabla^2 f(x_t)^{-1} \nabla f(x_t)$ (assuming that $\nabla^2 f(x_t)$ is invertible)

Assumptions ① x_t is close to x^*

② $\nabla^2 f(x^*)$ is invertible ($\lambda(\nabla^2 f(x^*)) > 0$)

③ $\nabla^2 f(x)$ is lipschitz continuous.

$\Rightarrow \lambda(\nabla^2 f(x))$ is ctz in x,

$\Rightarrow \lambda(\nabla^2 f(x_t)) > 0 \ \forall x_t$ close to x^*)

$x_t \bullet \ \ x^* \ \ >0$

$g(x^*) > 0$

g is ctz $\forall x$ close to x^*

$g(x) > 0$

Modified Newton's method

Fix: $\Delta, \alpha, \beta, \gamma$

Initialize: $x_0 \in \mathbb{R}^n$

Iterate: for [Equation]

If [Equation] is invertible and [Equation] (descent step)

[Equation]

Else

find [Equation] such that [Equation] (Δ is pre-fixed, say 0.5)

Set [Equation] and [Equation] according to Armijo's rule

[Equation]

end

$\lambda_{min}(\nabla^2 f(x_t) + \delta_t I) \geq \Delta$

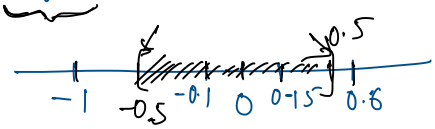
$d_t = -(\nabla^2 f(x_t) + \delta_t I)^{-1} \nabla f(x_t)$

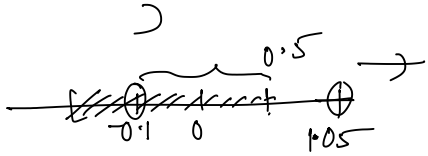
chosen to be fixed or using Armijo's rule

suppose $\nabla^2 f(x_t)$ is non invertible

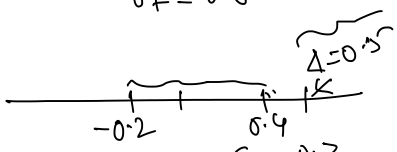
$\nabla^2 f(t) + \delta_t I$

$\Delta = 0.5$





$\delta_t = 0.6$



$\delta_t = 0.7$

Issue:

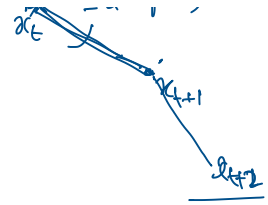
$x_{t+1} = x_t - \alpha \nabla f(x_t)$

$x_{t+1} - x_t = -\alpha \nabla f(x_t)$

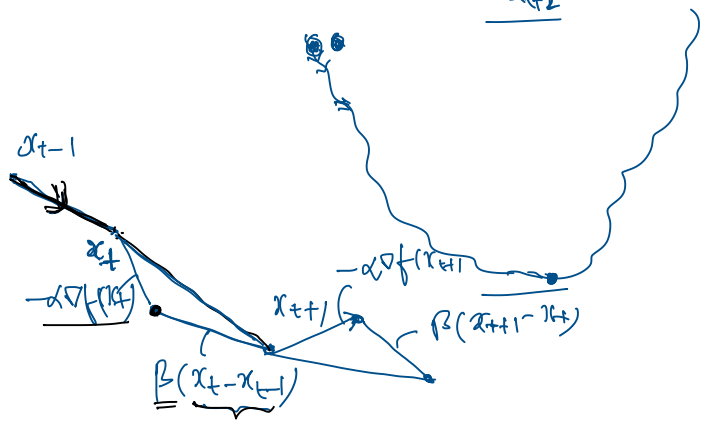
x_{t+1}

$\alpha \nabla f(x_t)$

$x_t - x_{t-1} = -\alpha \nabla f(x_{t-1})$



- Heavy ball methods
- Momentum methods



$$x_{t+1} = x_t - \alpha \nabla f(x_t) + \beta(x_t - x_{t-1})$$
$$= \underbrace{x_t + \beta(x_t - x_{t-1})}_{=: y_t} - \alpha \nabla f(x_t)$$
$$= y_t - \alpha \nabla f(x_t)$$

$$x_{t+1} - x_t = -\alpha \nabla f(x_t) + \beta(x_t - x_{t-1})$$

Polyak's heavy ball method:
Initialize: [Equation]
Iterate: for [Equation] where [Equation]
[Equation] where $\alpha > 0$

(∇f is Lipschitz cts with para L)

Theorem: If f is L-smooth and strongly convex with parameter m then using [Equation] and [Equation] ,
1. [Equation] ,
2. [Equation] .

$$\Rightarrow \|x_T - x^*\|^2 \le \left(\frac{\sqrt{L}-\sqrt{m}}{\sqrt{L}+\sqrt{m}}\right)^T \|x_0 - x^*\|^2$$

$(0.8)^{20}$

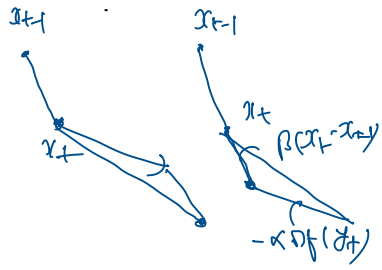
gradient descent f is L-smooth and strongly convex with parameter m

$$\|x_T - x^*\|^2 \le \left(\frac{\sqrt{L}-\sqrt{m}}{\sqrt{L}+\sqrt{m}}\right)^T \|x_0 - x^*\|^2$$

say 0.8, $T = 10$ $(0.8)^{10}$

$\sqrt{\frac{m}{L}}$ vs $\frac{m}{L}$

$$y_t = x_t + \beta_t (x_t - x_{t-1})$$
$$x_{t+1} = y_t - \alpha_t \nabla f(y_t)$$



Nesterov's accelerated gradient method:
Initialize: [Equation]
Iterate: for [Equation] where [Equation]
[Equation] where $\alpha > 0$

Theorem: If f is L-smooth and strongly convex with parameter m then using [Equation] and

[Equation] ,

1. [Equation] ,
2. [Equation] .

Suppose f is L-smooth (not necessarily convex)
gradient descent ensures

$$f(x_T) - f(x^*) \le \frac{C}{T} \qquad \left(O\left(\frac{1}{T}\right)\right)$$

Theorem: If [Equation] is [Equation] -smooth then using [Equation] and [Equation] , [Equation] .

$O\left(\frac{1}{T^2}\right)$

- In general these methods are not descent methods, e.g., [Equation] can be greater than [Equation] .

(here $f(x_{T+1}) \le f(x_T)$ is not guaranteed)

Comparison (objective: [Equation])

[Equation] (convex)	Number of iterations ([Equation])	Number of iterations ([Equation])
	~~Gradient descent~~	Accelerated gradient descent
Smooth (Lipschitz gradient)	[Equation]	[Equation]
Smooth and strongly convex	[Equation]	[Equation]

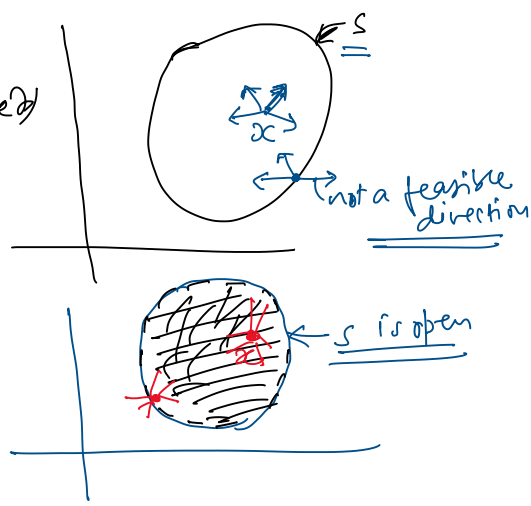
$\frac{1}{\epsilon} > \frac{1}{\sqrt{\epsilon}}$

$\sqrt{\frac{L}{m}} < \frac{L}{m}$

Constrained optimization: [Equation] , [Equation] continuously differentiable [Equation]

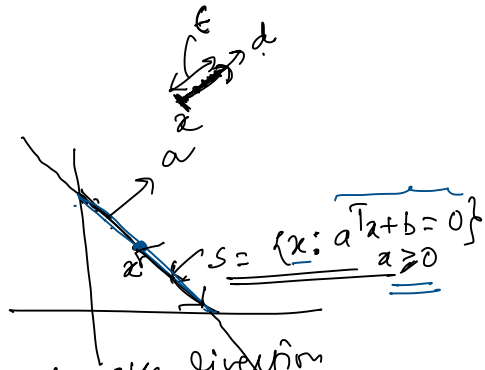
Earlier $\min_{x \in \mathbb{R}^n} f(x)$

(Typically, S is assumed to be closed)



Feasible directions: Given [Equation] , [Equation] is called a feasible direction at [Equation]

if there exists an [Equation] such that [Equation] for all [Equation] .



suppose d is a feasible direction

$\Rightarrow \exists$ an $\epsilon > 0$ such that $x + \alpha d \in S \quad \forall \alpha < \epsilon$

$$\Leftrightarrow a^T(x + \alpha d) + b = 0$$

$$\Leftrightarrow \underbrace{a^T x + b}_{=0} + \alpha a^T d = 0$$

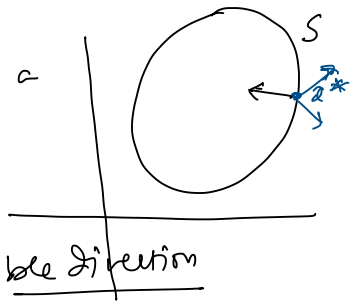
$$\Leftrightarrow a^T d = 0$$

$\min_{x \in S} f(x)$

x^* is a local minimum, d is a feasible direction $\Rightarrow \exists \epsilon > 0$ s.t.

$f(x^* + \alpha d) \geq f(x^*) \quad \forall \alpha \in (0, \epsilon]$

$\in S$ because d is feasible direction



$$\cancel{f(x^*)} + \alpha d^T \nabla f(x^*) \geq \cancel{f(x^*)}$$

$$\Leftrightarrow \nabla f(x^*)^T d \geq 0$$

First order necessary condition (of optimality)

Lemma: If [Equation] continuously differentiable, [Equation] is a local optimum and [Equation] is a feasible direction at [Equation] then [Equation] .

unconstrained prob. is a special case

x^* is local min

$$\Rightarrow \nabla f(x^*)^T \cdot \underbrace{\nabla f(x^*)}_{\text{a feasible dir}} \geq 0$$

$$\Rightarrow \|\nabla f(x^*)\|^2 \geq 0 \quad \text{— (A)}$$

But $\nabla f(x^*)^T(-\nabla f(x^*)) \geq 0$

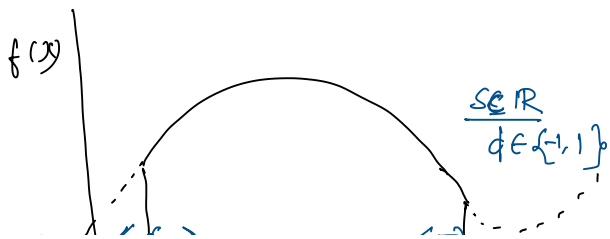
$$\Rightarrow \|\nabla f(x^*)\|^2 \leq 0 \quad \text{— (B)}$$

(A) + (B) $\Rightarrow \|\nabla f(x^*)\|^2 = 0 \Leftrightarrow \nabla f(x^*) = 0$

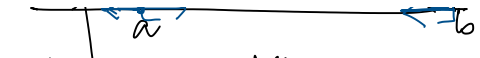
Example

$\min_{x \in [a,b]} f(x)$

Both $x = a$, b are local



minima



- there do not satisfy $f'(a)=0$ or $f'(b)=0$

$d=+1$ is the only feasible dir^n at a

$d=-1$ " " " " b

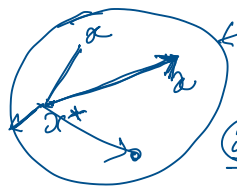
for a, necessary cond $f'(a)\cdot(+1)\geq 0$

for b, " " $f'(b)(-1)\geq 0$

both are satisfied

So, a and b meet Ist order necessary conditions

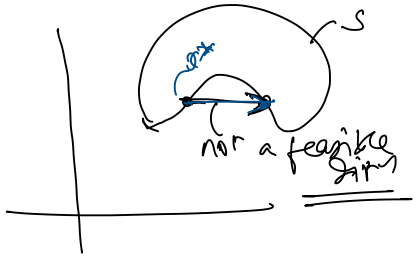
min $f(x)$
$x\in S$
where S is convex.



$\forall x\in S$ $(x-x^*)$ is a feasible dir^n

Necessary cond^n for x^* to be local minimum

$\nabla f(x^*)^T(x-x^*)\geq 0 \quad \forall x\in S$



Suppose f is also a convex fn

Also $\nabla f(x^*)^T(x-x^*)\geq 0 \quad \forall x\in S$ ✓

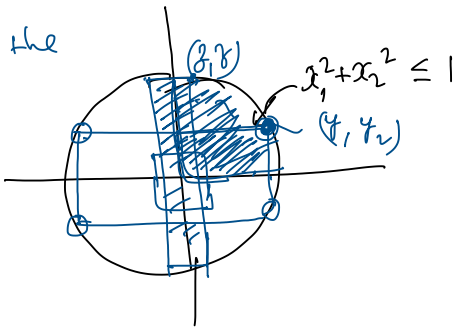
$f(x)\geq f(x^*)+\underbrace{\nabla f(x^*)^T(x-x^*)}_{\geq 0}$

$\Rightarrow f(x)\geq f(x^*) \Rightarrow x^*$ is global minimum.

Lemma: (a) ~~If~~ [Equation] is convex, [Equation] is continuously differentiable and [Equation] is a local
optimum then [Equation] for all [Equation] .
(b) In addition to the above assumptions, if [Equation] is also convex then [Equation] for all [Equation]
is also a sufficient condition for [Equation] (global minimum) to be an optimum.

Examples

(1) the largest area rectangle with inside the disc



min $-x_1x_2$

s.t. $\underbrace{x_1^2+x_2^2\leq 1}_{(S,\ x\geq 0)}$, $x\geq 0$

let (x_1^*,x_2^*) is the optimal sol^n

$$\nabla f(x) = \begin{bmatrix} -x_2 \\ -x_1 \end{bmatrix}$$

Necessary cond: $\nabla f(x^*)^T (x - x^*) \geq 0 \quad \forall\, x \in S$

$$\Rightarrow [x_2^* \;\; x_1^*] \begin{bmatrix} x_1 - x_1^* \\ x_2 - x_2^* \end{bmatrix} \leq 0$$

$$\Rightarrow x_1 x_2^* - x_1^* x_2^* + x_2 x_1^* - x_1^* x_2^* \leq 0$$

$$\Rightarrow \frac{x_1}{x_1^*} + \frac{x_2}{x_2^*} \leq 2$$

$$\Rightarrow \frac{1}{2}\left(\frac{x_1}{x_1^*} + \frac{x_2}{x_2^*}\right) \leq 1 \quad \text{—— } (*)$$

$$\nabla^2 f(x) = \underbrace{\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}}_{\text{neither PSD nor NSD}}$$

$\lambda_1 + \lambda_2 = 0$

$\lambda_1 \lambda_2 = -1$

(2nd order nec. cond. for unconstrained prob)

We argue that $(*)$ is a sufficient cond.

$$\underbrace{\sqrt{\frac{x_1}{x_1^*} \cdot \frac{x_2}{x_2^*}}}_{GM} \underset{(GM \leq AM)}{\leq} \underbrace{\frac{1}{2}\left(\frac{x_1}{x_1^*} + \frac{x_2}{x_2^*}\right)}_{AM} \underset{(*)}{\leq} 1$$

$$\Rightarrow \sqrt{x_1 x_2} \leq \sqrt{x_1^* x_2^*} \quad \forall\, x \in S$$

$$\Rightarrow x_1 x_2 \leq x_1^* x_2^* \quad \forall\, x \in S$$

$$\Rightarrow -x_1 x_2 \geq -x_1^* x_2^* \quad \forall\, x \in S$$

$\Rightarrow (x_1^*, x_2^*)$ is a global minimum

Suppose $x_1^* = 1/\sqrt{2}$, $x_2^* = 1/\sqrt{2}$

We argue that $\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$ is a global minimum

$\forall\, x \in S$

$$x_1 x_2 \leq \underbrace{x_1^2 + x_2^2}_{(AM \geq GM)^2} \leq \frac{1}{2}$$

$\Rightarrow x_1 x_2$ cannot exceed $\frac{1}{2}$ for any $x \in S$

But $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$

$\Rightarrow \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$ is a global minimum of $-x_1 x_2$

- there is no other global minimum. (Exercise)

Ex 2 (Optimization over a simplex) (next time)